Natalie Goldsworthy

natalie@goldsy.net | natgold11.github.io/ | linkedin.com/in/natalie-goldsworthy-h2p

EDUCATION

University of Pittsburgh (GPA: 3.85/4.0) **Expected May 2028**

BS in Computer Science, Data Science, and Digital Narrative & Interactive Design *Pittsburgh, PA*

- Minors: English Literature, Creative Writing
- Courses: Data Structures & Algorithms, Discrete Structures, Assembly & Computer Organization, Intro to System Software, Human Language Technologies, Software Engineering, Calculus I and II, Linear Algebra
- Activities: Kappa Theta Pi, Pitt Performance Collaborative, Computer Science Club, Pitt Ballet Club
- Leadership: Kappa Theta Pi - President (26-27), Vice President of Philanthropy (25-26); Performance Collaborative - Secretary/Treasurer (26-27)

WORK EXPERIENCE

Implementation and Game Designer **August 2026 - Present**

Entertainment Technology Center at Carnegie Mellon University *Pittsburgh, PA*

- Executed first iteration of multi-day augmented reality game featuring web-based puzzles built with JavaScript.
- Handled web development, web hosting through Firebase, and management of the MongoDB database.
- Contributed to a Gemini-powered chatbot, refining purpose and conversational logic for player interactions.
- Facilitated communication across a 10-person cross-institutional team, managing scheduling and deliverables.
- Redesigned and built full second iteration of the game elements based on outcomes from a 4-day pilot program.

Full Stack Software Engineering Intern **June 2026 - August 2026**

CGI *Pittsburgh, PA*

- Integrated a RAG-powered knowledge base into an existing ticketing dashboard for a leading regional bank.
- Architected a PostgreSQL and pgvector database to store and query document and ticket embeddings.
- Built a document ingestion pipeline with FastAPI and Docling to parse, chunk, and summarize files using Anthropic's Claude API.
- Developed a React dashboard with interactive tiles and side panels to streamline document information access.
- Containerized the full application stack using Docker and Rancher Desktop for consistent local development.

RESEARCH

Technical Design Team Member **January 2026 - May 2026**

HeritageRoots *University of Pittsburgh*

- Built educational web platform to teach K-12 students about indigenous cultures through game-like experiences.
- Developed 360-degree image exploration using the Pannellum API, HTML, CSS, and JavaScript.
- Worked alongside team of developers to complete weekly deliverables and efficiently delegate work.
- Prioritized regular and effective communication between team members and supervising professor.

PROJECTS

Hunger Games: Survive the Arena (Unity, C#)

Link: natality.itch.io/hunger-games-survive-the-arena

- Created a 2D roguelike shooter inspired by the Hunger Games franchise using the Unity game engine.
- Developed core systems including movement, combat, enemy AI, and a modular upgrade system.
- Implemented a randomized run-based progression system balancing difficulty scaling with player upgrades

SKILLS

Languages: Java, Python, JavaScript, SQL, C#, C, MIPS Assembly, HTML, CSS

Libraries/Frameworks/APIs: React, Pandas, Flask, FastAPI, Docling, LangChain, Hugging Face, Anthropic API

Databases: PostgreSQL, pgvector, SQLite

Tools: Docker, Rancher Desktop, Swagger UI, pgAdmin, Git/GitHub, Unity, VS Code, Jira, Confluence, Adobe Suite